

ICLR 2025 | Why Generative Models Memorize — and How Geometry Explains It

Layer 6 AI's Manifold Memorization Hypothesis measures memorization with local intrinsic dimension, all the way up to Stable Diffusion

Type "a Van Gogh starry night" into a text-to-image model and you get back a picture. But did the model learn Van Gogh's style, or did it simply regurgitate one of its training images? As diffusion models grow more capable, this stops being an academic curiosity. A model that memorizes and reproduces its training data can leak private information and reproduce copyrighted works, exposing whoever built it to genuine legal liability and risk.

The trouble is that the field has lacked a single, formal ruler for the question "how memorized is this sample?" Most existing definitions lean on pixel-space distance to the nearest training image. That misses the subtler cases — reconstructive memorization, where the layout or composition is copied even though the image is not a pixel-perfect duplicate — and it does not scale to LAION-2B and Stable Diffusion, where computing distances against the whole training set is hopeless.

This ICLR 2025 paper from Layer 6 AI offers a clean answer: memorization is fundamentally a geometric phenomenon. The authors introduce the Manifold Memorization Hypothesis (MMH), which characterizes how memorized a point is through a single measurable geometric quantity — its local intrinsic dimension (LID). That framing lets them cleanly separate two very different causes of memorization, and it yields detection and mitigation tools that scale all the way to Stable Diffusion.

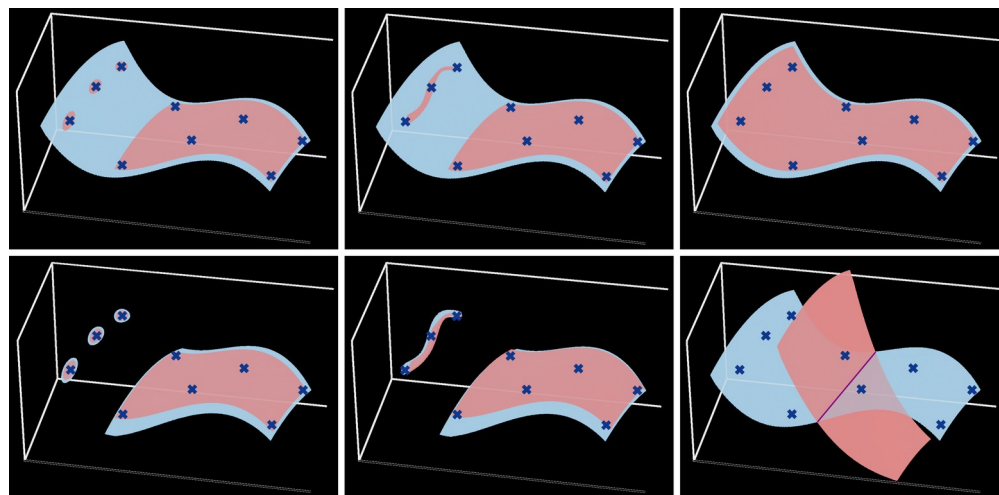


Figure 1. Local intrinsic dimension (LID) as a lens on memorization. The ground-truth manifold is light blue, training samples are crosses, and the learned model manifold is red. In (a) and (d) the model collapses to 0-dimensional point masses around datapoints — memorization; (a) is overfitting-driven ($LID_0(x) < LID^*(x)$), (d) is data-driven (the ground-truth LID is itself small). (b) and (e) memorize a 1-dimensional submanifold. Only (c), which recovers the true 2-dimensional manifold, generalizes and avoids both. (f) is a poorly fit model where LID and memorization are not meaningfully related.

Paper: <https://arxiv.org/abs/2411.00113>
https://github.com/layer6ai-labs/dgm_geometry

Code:

Why reach for geometry

The manifold hypothesis holds that realistic data lives on a low-dimensional manifold embedded in a high-dimensional space. The authors argue this geometry is exactly the right language for memorization. Purely probabilistic definitions demand the full training set and enormous sample counts to estimate density — intractable at LAION scale. A geometric framing instead ties memorization to a quantity for which practical estimators already exist, even for a model as large as Stable Diffusion.

LID(x) is best read as the number of degrees of freedom a point x has on its manifold. Think of dimensions as constraints: in a d -dimensional space, each active constraint removes one degree of freedom, so ℓ constraints give $\text{LID}(x) = d - \ell$. A training image that the model reproduces pixel-for-pixel has essentially no freedom left — it sits on a 0-dimensional point mass, which is memorization in its most extreme form.

One word, two very different causes

The core move of the MMH is to compare $\text{LID}_\theta(x)$ on the model's learned manifold M_θ against $\text{LID}^*(x)$ on the ground-truth manifold M^* . How those two quantities relate carves memorization into two causes that look identical from the outside but are anything but — and this distinction is the paper's sharpest idea.

The first is overfitting-driven memorization (OD-Mem): the model's local dimension is lower than the truth ($\text{LID}_\theta(x) < \text{LID}^*(x)$). The model has crushed a region that should admit many outputs down to a single point — a genuine failure to generalize. The second is data-driven memorization (DD-Mem): the model is not wrong at all ($\text{LID}_\theta(x) = \text{LID}^*(x)$); the point's true dimension is simply low to begin with. Picture a highly specific prompt like the title of a famous artwork, for which few plausible outputs exist. The crucial consequence: DD-Mem cannot be caught by comparing train and test likelihoods, because nothing was overfit.

To test the framework you first need a setting where the true dimension is known. The authors train a diffusion model on a two-dimensional von Mises mixture: because the ground-truth manifold is known, LID^* can be computed exactly and checked against the model's estimated LID_θ .

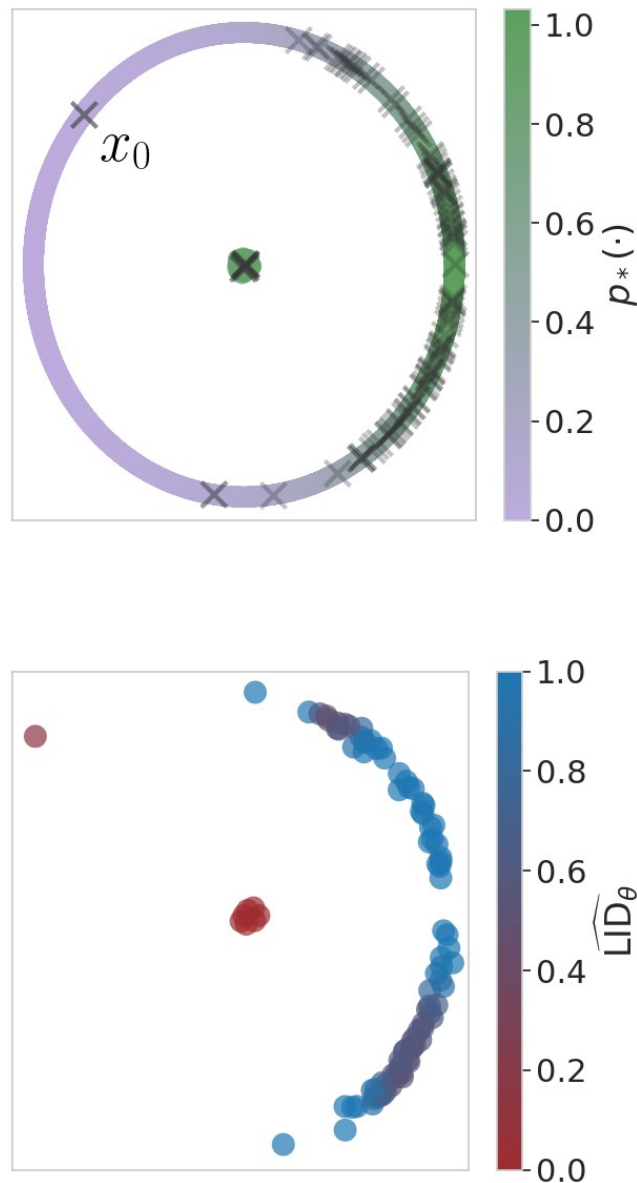


Figure 4. A diffusion model trained on a 2D von Mises mixture. Top: the ground-truth manifold and its distribution (a ring plus a central point mass). Bottom: model-generated samples colored by their estimated LID $_{\theta}$ — samples on the ring (1-dimensional) get high LID, samples collapsing to the central point (0-dimensional) get low LID, matching the true geometry.

CIFAR10: what the two kinds of memorization look like

One rung up the ladder, the authors train StyleGAN2-ADA and an iDDPM diffusion model on CIFAR10. Here you can see both exact memorization (a generated image nearly pixel-identical to a training image) and reconstructive memorization (the subject and layout are copied while the details differ) — the latter being exactly the case pixel distance tends to miss. Low-LID $_{\theta}$ samples of either kind are reliably flagged by the same ruler.

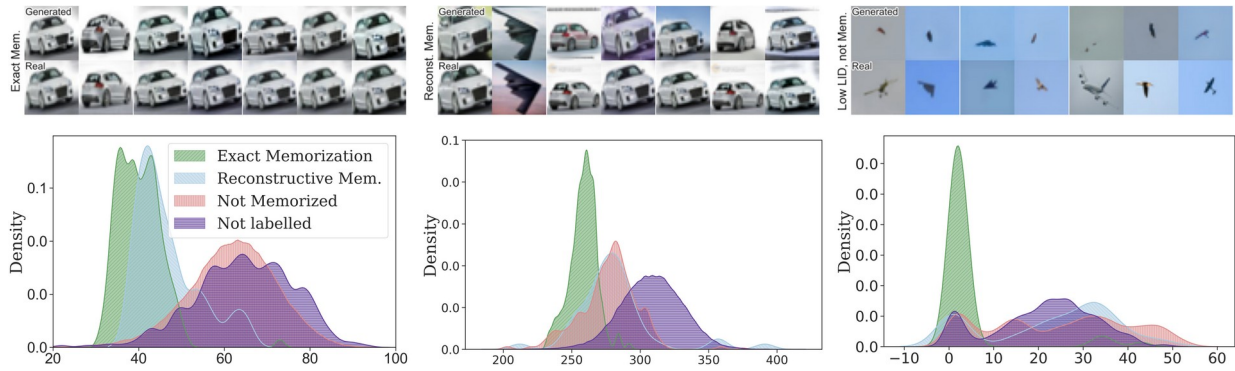


Figure 5. Visualizing OD-Mem and DD-Mem on CIFAR10 (StyleGAN2-ADA and iDDPM). Top row: exactly memorized (left) and reconstructively memorized (middle) generated samples beside their matched CIFAR10 images, plus low-LID but non-memorized samples (right). Bottom row: density histograms of each memorization metric — memorized samples take clearly lower LID and separate cleanly from non-memorized ones.

Scaling to Stable Diffusion: detecting memorization without the prompt

The real test is Stable Diffusion v1.5. Because true LID^* cannot be estimated at this scale, the authors switch to three scalable, model-side proxies: unconditional $LID\theta$ (estimated with the FLIPD algorithm), conditional $LID\theta(\cdot|c)$ given the prompt, and the classifier-free guidance (CFG) vector norm used by prior work. They also prove an intuitive inequality — conditioning can only lower the local dimension ($LID^*(x_0|c) \leq LID^*(x_0)$) — which explains why more specific prompts are more prone to triggering the memorization behavior.

Comparing 86 verbatim-memorized LAION images against several thousand non-memorized ones (2000 from LAION-Aesthetics, 2000 from COCO, and all 251 from Tuxemon), all three proxies point the same way: memorized images consistently get low $LID\theta$ and high CFG vector norm. More striking is a genuinely new capability — unconditional $LID\theta$ flags memorized training images without needing their prompts at all, whereas the older CFG-vector-norm method requires the caption in hand before it can say anything.

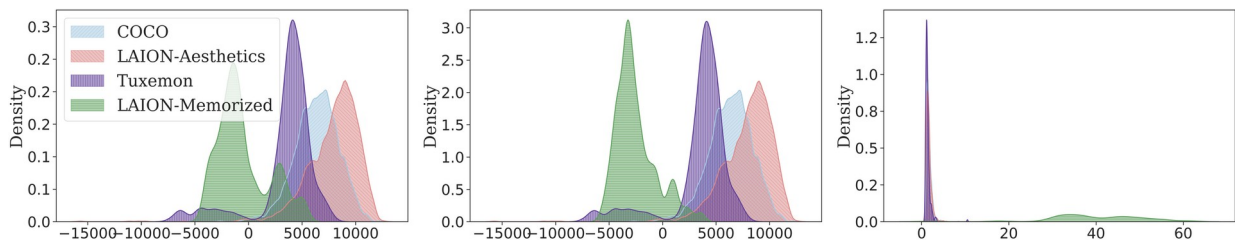


Figure 6. Density histograms for each memorization metric ($LID\theta$, conditional $LID\theta$, and CFG vector norm) across datasets on Stable Diffusion. The green LAION-Memorized distribution separates clearly from the non-memorized COCO, LAION-Aesthetics, and Tuxemon data, showing these geometric quantities can pick out memorized images.

The three proxies each make a different trade-off, and the table below lays them side by side for an at-a-glance comparison.

Table 1. Three memorization-detection proxies that remain tractable at Stable Diffusion scale.

Proxy	Geometric meaning	On memorized images	Needs the prompt?
Unconditional LID θ (FLIPD)	Local dimension of the unconditional manifold	Low	No
Conditional LID $\theta(\cdot c)$	Local dimension given the prompt	Low	Yes
CFG vector norm	Magnitude of the guidance vector	High	Yes

Detection and mitigation: attributing memorization to individual tokens

If memorization is geometric, the way to mitigate it follows naturally: steer generation toward regions of higher local dimension. The authors backpropagate a differentiable memorization metric to attribute memorization to each token in the prompt, identify the words that most drive the dimension down, and use GPT-4 to rephrase them. Compared with rephrasing randomly chosen tokens, attribution-based selection preserves higher image quality at the same reduction in memorization.

Sweeping the number of perturbed tokens k over $\{1, 2, 3, 4, 6, 8\}$ traces the trade-off between memorization (SSCD similarity, lower is better) and fidelity (CLIP score, higher is better): perturb more tokens and memorization falls, but so does fidelity. Two new attribution metrics — one built on the CFG-adjusted score norm, one on FLIPD — work about as well as the original CFG-vector-norm metric of Wen et al. (2023), and every attribution-based variant beats random token perturbation.

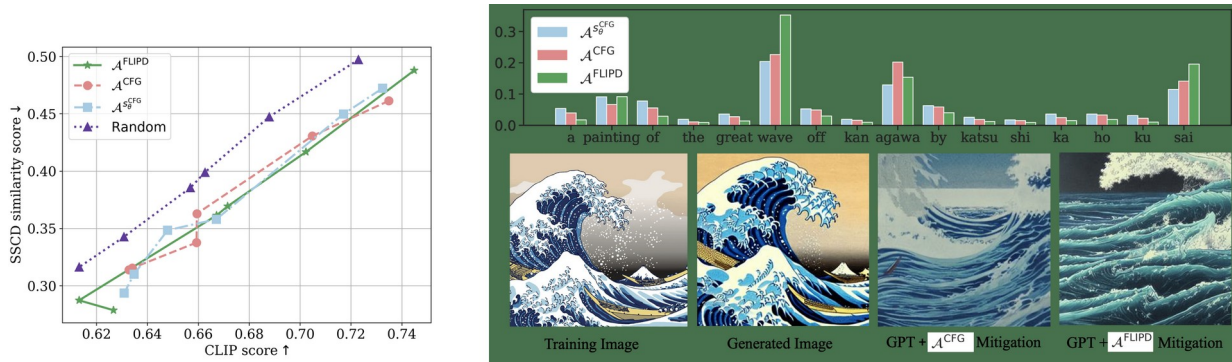


Figure 7. Using token attributions to diagnose and mitigate memorization. Left: each attribution metric (A^{FLIPD} , A^{CFG} , $A^{s_{\theta} CFG}$) reaches a lower SSCD similarity than Random at equal CLIP score. Right: attribution scores for a memorized prompt across three methods, alongside the training image, the model's generation, and two mitigated results that no longer reproduce The Great Wave verbatim.

The takeaway: memorization is geometric

The single idea to carry away is that memorization in generative models is geometric. When a model's local intrinsic dimension around a point is too small, that point is memorized — whether the cause is overfitting or the data itself being inherently simple. Because LID can be estimated at scale, you can both detect memorization (even without the prompt) and mitigate it by steering generation toward higher-dimensional regions of the manifold.

For anyone shipping a generative model, the practical payoff is twofold. Because unconditional LID θ needs no caption, a model owner can audit their own training set for memorized images directly, a check that was previously blocked whenever the offending prompt was unavailable. And because the same geometry that flags memorization also points the way out of it, the token-attribution-and-rephrasing loop becomes a lightweight, sample-time guardrail that leaves image quality largely intact.

The framework is honest about its limits. True manifold dimension LID* cannot be estimated directly on large models, so detection in practice relies on the model-side proxies; mitigation trades away some fidelity and has to be tuned by the number of tokens perturbed; and the large-scale validation rests on the 86 LAION images already flagged as verbatim-memorized, a still-limited slice. But the value of the MMH is that it turns a long-fuzzy, much-debated phenomenon into a measurable geometric quantity — usually the first real step toward making progress on it.

Zooming out, the more durable contribution may be the reframing itself. By recasting a contested, hard-to-pin-down behavior as a single geometric quantity, the paper hands the community a shared vocabulary and a measurable target. Future work can now argue about better LID estimators, tighter conditional bounds, or new mitigation strategies on common ground, instead of talking past one another about what memorization even means.